

741 Final Assignment

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Problem 2

Simulate data from $y_i = x_i^T \beta_0 + \epsilon_i$ where $p = 30, n = 50, \epsilon_i \sim N(0, 1)$. True β_0 has 10 components between 0.5 and 1, 20 components between 0 and 0.2. Draw all entries in design matrix iid from $N(0, 1)$.

```
library(glmnet) # for fitting Lasso regression model

## Loading required package: Matrix

## Loaded glmnet 4.1-8

trials <- 100
p <- 30
n <- 50
test_errors <- vector()
test_errorsLass <- vector()
predictions <- matrix(0,nrow=n,ncol=trials) # 50x100 matrices for fitted y values
predictionsLass <- matrix(0,nrow=n,ncol=trials)

# generating data and true coefficients ONLY ONCE
data <- matrix(rnorm(n=n*p,mean=0,sd=1),nrow=n) # generate data
small_comp <- runif(20,0,0.2) # generate true coefficients, small and large
large_comp <- runif(10,0.5,1)
coeff <- c(small_comp,large_comp) # combine true coefficients
true_coeff <- as.matrix(sample(coeff,replace=F)) # randomize order of true coefficient
ytest <- data%%true_coeff

for (i in 1:trials){
  ## (1)
  errors <- rnorm(n=n,mean=0,sd=1) # to add random noise, so fit is not perfect
  y <- ytest + errors # outcome vector

  ## (2)
  ls_model <- lm(y~0+X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+
                X11+X12+X13+X14+X15+X16+X17+X18+X19+X20+
                X21+X22+X23+X24+X25+X26+X27+X28+X29+X30,data=data.frame(data))
  betahat <- as.matrix(ls_model$coefficients) # extract coefficients

  cross_val <- cv.glmnet(data,y, alpha=1) # cross-val for lambda, alpha=1 for lasso
  lamb_opt <- (cross_val$lambda.min) # save optimal lambda
```

```
lass_mod <- glmnet(data,y,alpha=1,lambda=lamb_opt,intercept=FALSE) # fit model
betahatLass <- as.matrix(coef(lass_mod))[2:31,]

## (3)
yhat <- data %*% betahat # predicted outcome vector, fitted values
predictions[,i] <- yhat # ith column of predictions set to fitted values
test_errors[i] <- mean((y-yhat)^2) + var(errors) # MSE

yhatLass <- data %*% betahatLass # same but with lasso
predictionsLass[,i] <- yhatLass
test_errorsLass[i] <- mean((y-yhatLass)^2) + var(errors)
}

## (4)
exp_error <- mean(test_errors)
cat('The estimated expected test error over 100 MLR trials was',exp_error,'\n')
```

```
## The estimated expected test error over 100 MLR trials was 1.475246
```

```
exp_errorLass <- mean(test_errorsLass)
cat('The estimated expected test error over 100 Lasso trials was',exp_errorLass,'\n')
```

```
## The estimated expected test error over 100 Lasso trials was 1.636198
```

Expected test error = $\sigma^2 + Bias^2 + Variance$

```
## (5)
exp_MLR <- apply(predictions,1,mean) # mean MLR prediction for each obs
exp_Lass <- apply(predictionsLass,1,mean) # mean lasso prediction for each obs
bias_MLR <- mean(exp_MLR - y)
bias_Lass <- mean(exp_Lass - y)
var_MLR <- mean(apply(predictions,1,var)) # mean variance of prediction for each obs
var_Lass <- mean(apply(predictionsLass,1,var))
results_summary <- data.frame(Model=c('Least Squares','Lasso'),
                              MSE = c(exp_error,exp_errorLass),
                              Bias_Squared = c(bias_MLR^2,bias_Lass^2),
                              Variance = c(var_MLR,var_Lass),
                              ErrorVar=c(var(errors),var(errors)))

results_summary
```

```
##           Model      MSE Bias_Squared  Variance  ErrorVar
## 1 Least Squares 1.475246  0.02761660 0.6279823 0.7514653
## 2           Lasso 1.636198  0.02113203 0.4653101 0.7514653
```

Comments:

The Lasso results may be explored in more detail to see which parameters were shrunk to zero. Additionally, plots of the cross validation function can be easily made to visualize how choice of λ affects the error of the model.

Both models were fitted without intercept for the sake of simplicity.

The result $\text{MSE} = \text{Bias}^2 + \text{Variance} + 1$ is expected to hold more accurately across a larger number of trials.

Lasso has a slightly higher MSE and bias, but lower variance. Regardless, its utility is in shrinking parameters to zero to reduce model complexity.